Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 17380670409 58-Chapter 5_unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

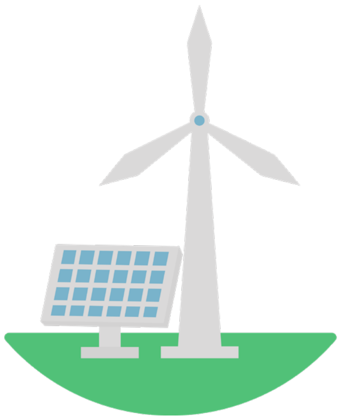
UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

TWINKLE TUNES: EUN'S MUSICAL DISCOVERY

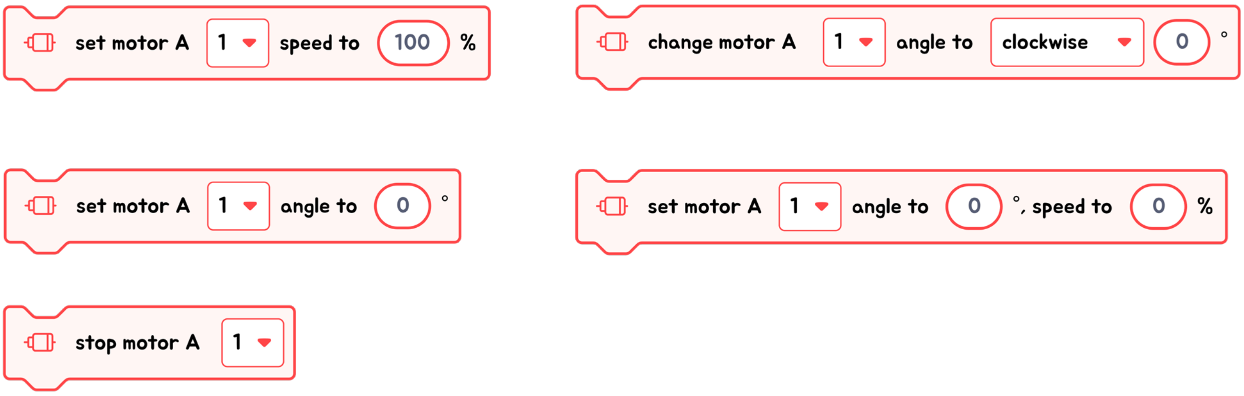
One sunny afternoon, seventh grader Hugh excitedly proposed a project to his friends, Cathy and Dawn: "Let's build a solar windmill!" Inspired by their recent lessons on renewable energy, the trio was eager to see these principles in action. Gathering in Hugh's backyard, they pooled their resources—small motors and solar panel from Cathy's old kit.

As they laid out the materials, Hugh suggested, "What if our windmill could use both wind and solar power?" His idea was to make the windmill blades rotate with wind energy while using solar power to light it up. Cathy and Dawn loved the hybrid concept, which perfectly demonstrated the synergy between different renewable energies. Filled with enthusiasm, they began sketching designs, ready to bring their innovative solar windmill to life.



Their completed solar windmill not only functioned as a brilliant school project but also sparked interest within their community. Hugh, Cathy, and Dawn showcased their creation at the local community center, explaining how renewable energy could promote environmental sustainability. They inspired neighbors to consider how small-scale renewable energy projects could be implemented in their homes.

ABOUT MOTOR A MODULE

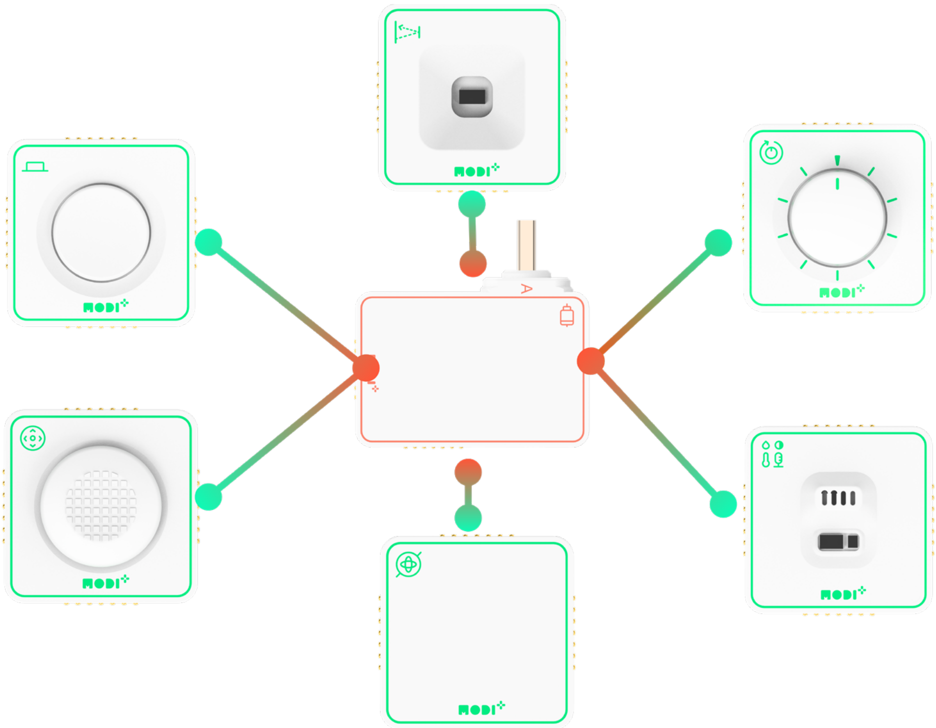


Motor A features five different blocks in the block area. These blocks allow you to customize your creations by setting the motor's angle, spinning direction, and speed.

INTERACTIONS WITH OTHER MODULES

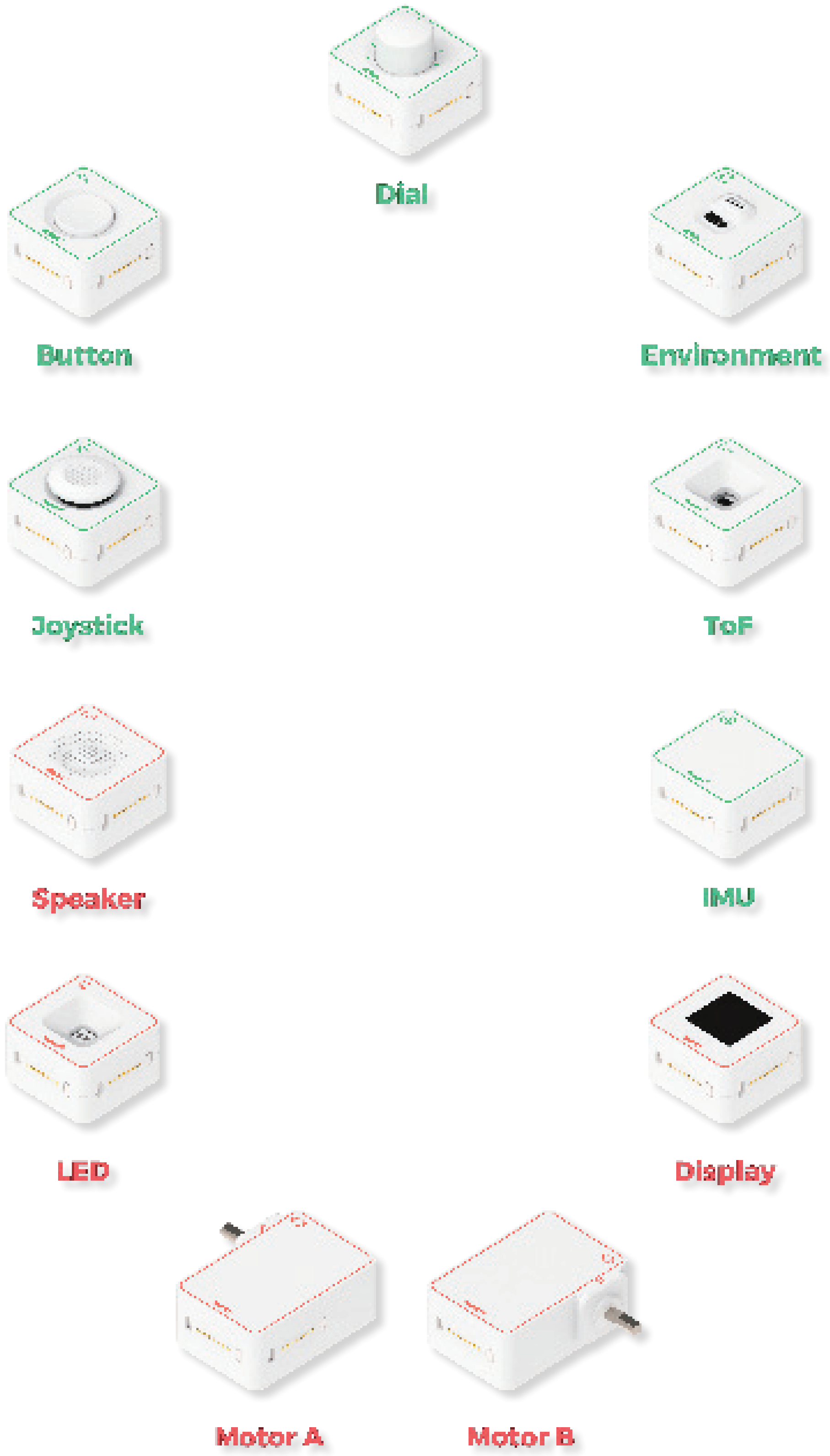
As an output module, Motor A can interact with all input modules, showing a unique response to each one. Explore these interactions by connecting Motor A to different input modules!

INTERACTION BETWEEN LED AND OTHER MODULES

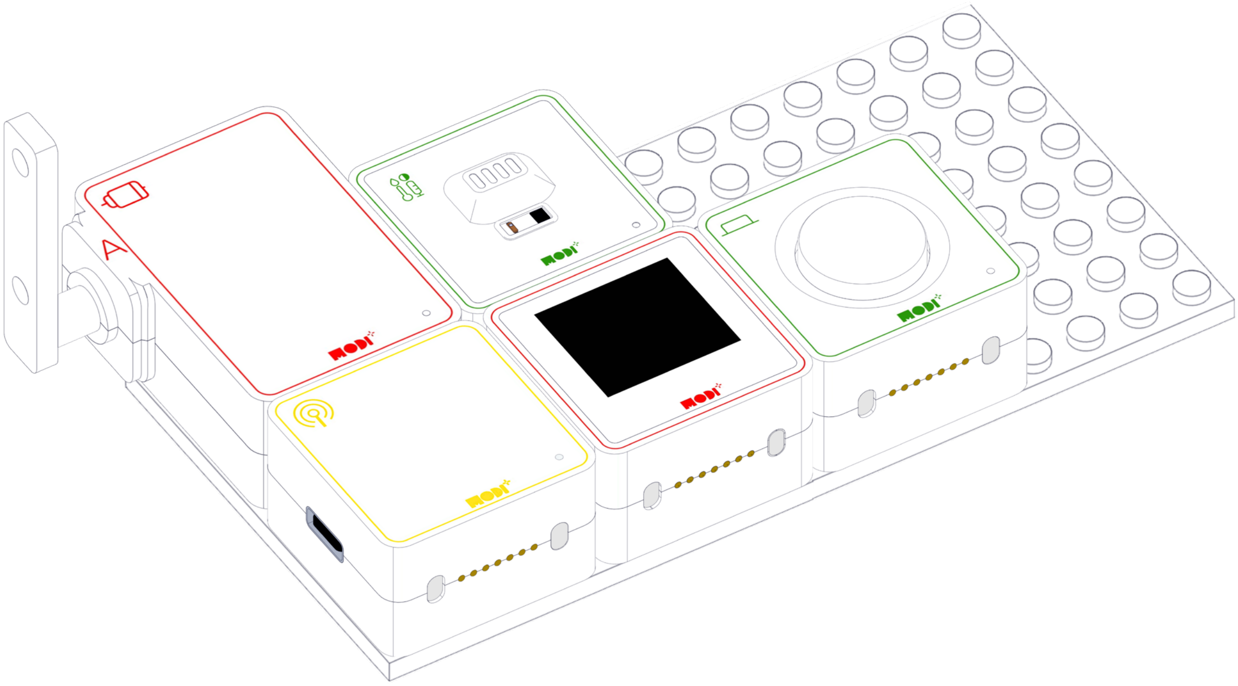


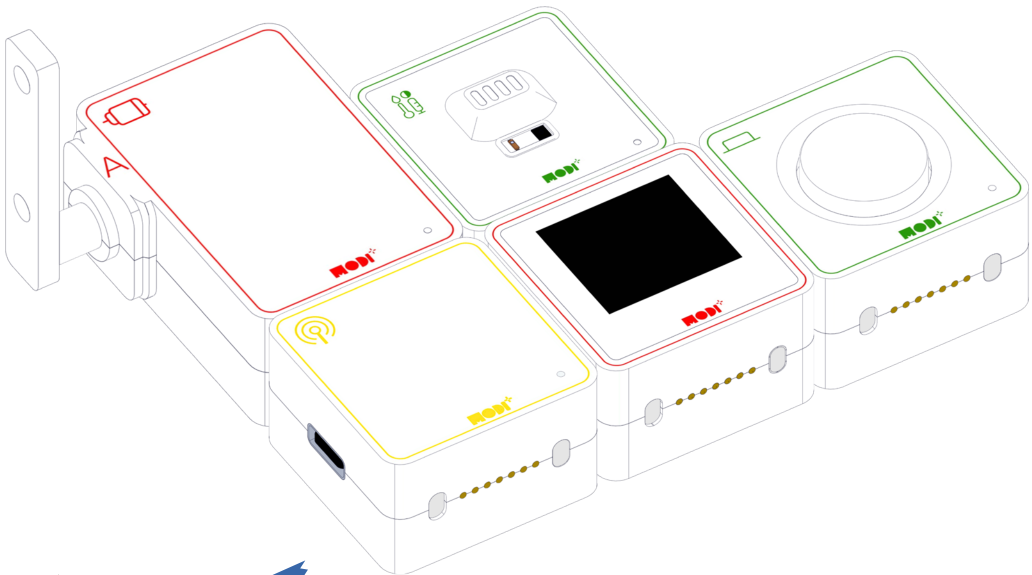
Required Modules & Brainstorming

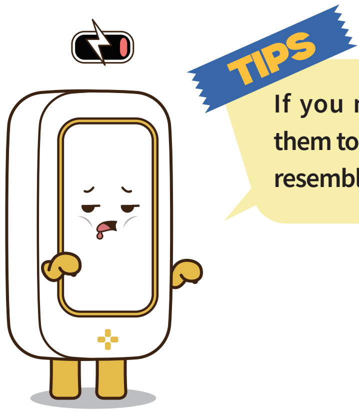
Based on the story, which MODI modules do you need for this creation?



HOW DOES THE CREATION LOOK LIKE?



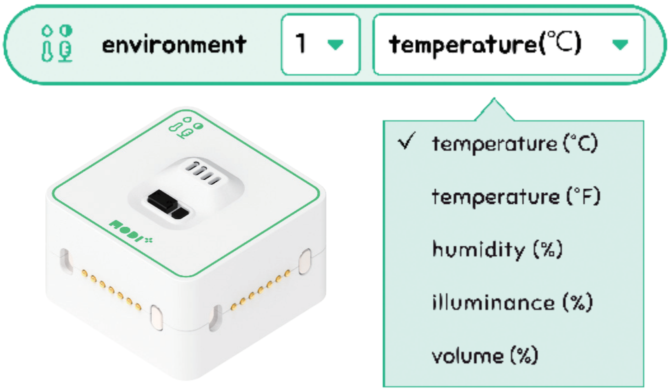




If you make wings and attach them to the I-horn, it will perfectly resemble a windmill.

Module Functionality

Let's check the details of the Environment module below for this project.

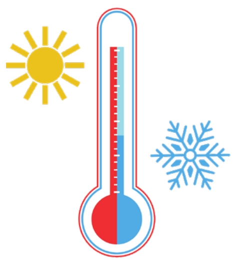


ENVIRONMENT

- **Temperature:** Measures how hot or cold thessurroundings are.
- **Humidity:** Checks the amount of moisture in the air.
- **Illuminance:** Gauges the brightness of the light around it.
- **Volume:** Detects the level of sound in the environment.

The Environment Module is a critical tool for monitoring conditions that impact renewable energy sources. It tracks three crucial environmental factors:

1. TEMPERATURE MEASUREMENT



This function measures the ambient temperature, which can affect the efficiency of solar panels and other energy systems. Maintaining an optimal temperature range is crucial for maximizing energy production.

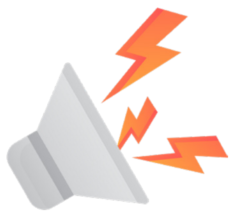
2. HUMIDITY MEASUREMENT

Humidity levels can influence the performance of renewable energy equipment. This part of the module checks the moisture in the air, which is essential for optimizing conditions for energy generation.

3. LIGHT MEASUREMENT

Light intensity directly impacts the output of solar energy systems. This device ensures there is sufficient sunlight for solar panels to operate effectively, crucial for areas relying on solar power.

4. NOISE MEASUREMENT



While typically less directly related to renewable energy, understanding noise levels can be important for the placement and operation of equipment like wind turbines, which may be sensitive to vibration or need to adhere to local noise regulations.